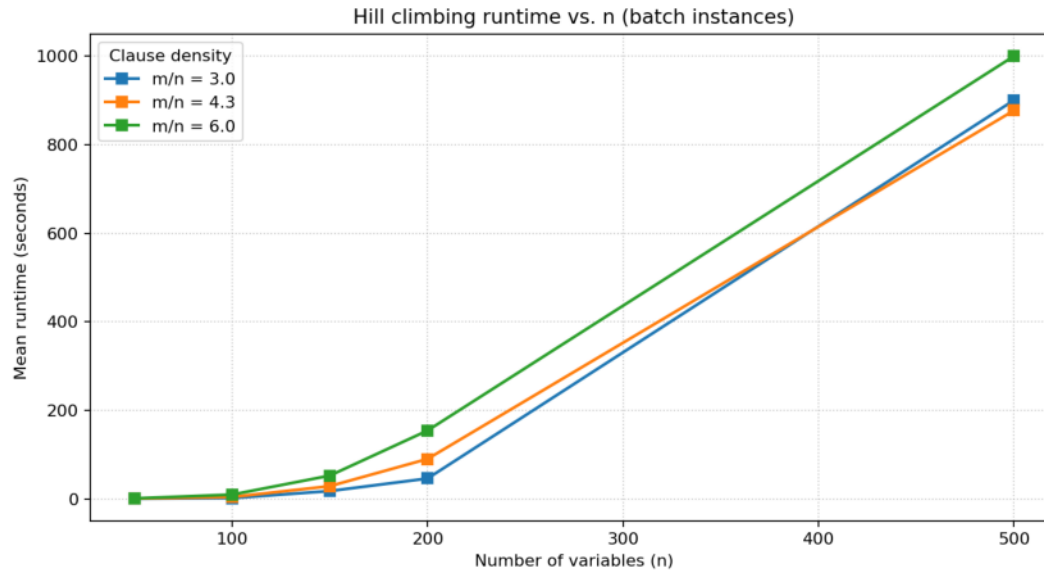


3-SAT Hill Climbing — Results Summary

Batch runtime vs. number of variables (by clause density)

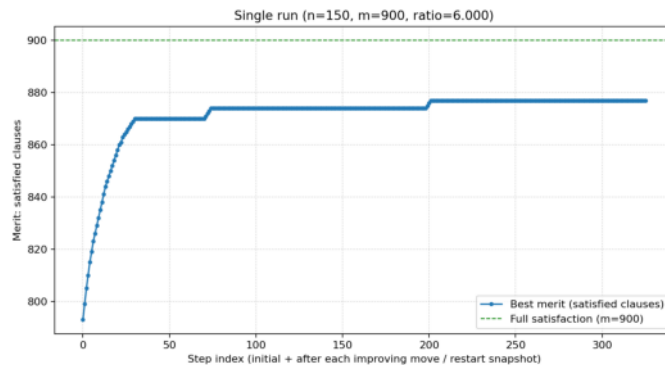


As n increases, mean runtime rises sharply. Higher clause density (m/n) is harder: $m/n = 6.0$ tends to be the slowest, followed by 4.3, then 3.0. This reflects a tighter search landscape with fewer easy improvements at higher densities.

Notes:

- Batch plot: runtime grows with n ; higher m/n increases difficulty.
- Single run: hill climbing plus random restarts; plateaus imply local optima.
- Restarts help escape local maxima by re-sampling assignments.

Convergence (merit over improving moves/restarts)



Rapid early gains followed by plateaus indicate local optima. Occasional jumps show successful restarts that discover better basins. Example shown reaches ~877/900 clauses at $n=150$, $m=900$ (ratio=6.0) – strong but not fully satisfying all clauses.